

Year 7 · Chapter 10 · Analysing data: answers (teacher copy)

10.01 Interpreting graphs (booklet p. 3)

WE DO examples (teacher)

1. Ex 1a: Gold Coast
2. Ex 1b: Uluru
3. Ex 2a: 15–24 years
4. Ex 2b: 12%
5. Ex 3a: Jan and Feb, 18°C
6. Ex 3b: it falls steadily to a low in July

Easy · Set A

1. Gold Coast
2. Uluru
3. 30
4. true (8)

Easy · Set A2 (quick drill)

1. 15
2. 50
3. 28
4. 12
5. 25
6. 20

Easy · Set B

1. 15–24 years
2. 12%
3. 16.5%
4. e.g. working-age adults and migrants

Medium · Set C

1. Jan and Feb, 18°C
2. e.g. Jan and Feb (or May and Sep)
3. October
4. July (3°C)
5. May–June or June–July (2°C)
6. it falls steadily to a low in July

Challenging · Set D

1. 22%
2. Anglican
3. No religion
4. e.g. more Christians and fewer “no religion”

Skill drill 1 (Easy basics)

- | | | |
|-----------|-------------|----------|
| 1. 13 | 10. 16°C | 19. 150 |
| 2. 8 | 11. January | 20. 15 |
| 3. 6 | 12. June | 21. 37.5 |
| 4. 15 | 13. 11°C | 22. 55 |
| 5. 6 | 14. 16°C | 23. 85 |
| 6. Tennis | 15. 47.5 | 24. 18 |
| 7. 48 | 16. 32.5 | 25. 130 |
| 8. 2 | 17. 11 | 26. 7 |
| 9. 27°C | 18. 170 | |

Skill drill 2 (Medium basics)

- | | | |
|----------|---------------------|-----------|
| 1. 13/48 | 10. it gets colder | 15. 95% |
| 2. 8/48 | (decreases) | 16. 96% |
| 3. 6/48 | 11. it gets warmer | 17. 16% |
| 4. 27.1% | (increases) | 18. 72.5% |
| 5. 16.7% | 12. southern: it is | 19. 35% |
| 6. 12.5% | coldest in | 20. 80% |
| 7. 9.6 | June–August | 21. 57% |
| 8. 5 | 13. 34% | 22. 84% |
| 9. 3 | 14. 67.5% | 23. 10% |

24. 55%

Extension

1. 76.5°
2. Gold Coast 6 cm, Uluru 3.4 cm, Snowy 3 cm, Kangaroo Is. 2 cm, Phillip Is. 1.6 cm

Exit ticket

1. Level 1: Phillip Island
2. Level 2: 11°C
3. Level 3: 8%

10.02 Misleading graphs (booklet p. 11)

WE DO examples (teacher)

1. Ex 1a: the vertical axis starts at \$2 million, not 0
2. Ex 1b: profits grew much faster than they did
3. Ex 2a: no
4. Ex 2b: 51 and 56
5. Ex 3a: uneven scale, pictures, no labels, missing data
6. Ex 3b: to impress investors

Easy · Set A

1. axis does not start at 0
2. huge growth
3. \$2.2 million
4. \$1.8 million

Easy · Set A2 (quick drill)

1. M
2. F
3. M
4. M
5. M
6. F

Easy · Set B

1. 50
2. 51
3. 58
4. no

Medium · Set C

1. line graph from 0 to 4 with even scale

Challenging · Set D

1. the area is 4 times bigger, not 2
2. you do not know what is shown
3. the steps are not equal
4. own answer

Skill drill 1 (Easy basics)

- | | | |
|------|------------------|--------------------|
| 1. M | 9. F | 17. missing labels |
| 2. F | 10. F | 18. scale |
| 3. F | 11. M | 19. picture size |
| 4. M | 12. M | 20. axis start |
| 5. F | 13. F | 21. missing labels |
| 6. M | 14. M | 22. scale |
| 7. M | 15. axis start | |
| 8. M | 16. picture size | |

Skill drill 2 (Medium basics)

- | | | |
|--------------------------------------|--------------------------------------|---------------------------------------|
| 1. small differences look very large | 2. the missing months hide the falls | 3. the change looks faster than it is |
|--------------------------------------|--------------------------------------|---------------------------------------|

- | | | |
|--|--|-------------------------------------|
| 4. its area is 4 times, so it looks 4 times as big | 6. the second column looks 3 times as tall, but sales only rose about 2% | 9. looks 2 times; really 1.0 times |
| 5. the front sector looks bigger than it is | 7. looks 4 times; really 1.1 times | 10. looks 5 times; really 1.4 times |
| | 8. looks 5 times; really 1.1 times | 11. looks 4 times; really 1.3 times |
| | | 12. looks 3 times; really 1.0 times |

Extension

- the axis starts at 2, so heights above 2 are compared (0.2 vs 2.0)
- e.g. axis from 0 to 20

Exit ticket

- Level 1: e.g. the axis does not start at 0
- Level 2: small differences look huge
- Level 3: start the vertical axis at 0 with an even scale

10.03 Dot plots (booklet p. 19)

WE DO examples (teacher)

- Ex 1a: 23°C
- Ex 1b: 15°C
- Ex 2a: 4
- Ex 2b: $\frac{1}{15}$
- Ex 3a: dot plot
- Ex 3b: most days between 20°C and 26°C, peak at 23°C

Easy · Set A

- 23°C
- 15°C
- 4
- $\frac{1}{15}$

Easy · Set A2 (quick drill)

- 3
- 8
- 2
- 12
- 7
- 3

Easy · Set B

- 16
- 1
- 7
- 8

Medium · Set C

- dot plot
- dot plot

Challenging · Set D

- dot plot
- mode 0, outlier 15; it makes the mean bigger

Skill drill 1 (Easy basics)

- | | | |
|------|-------|--------|
| 1. 1 | 10. 4 | 19. 9 |
| 2. 2 | 11. 7 | 20. 8 |
| 3. 8 | 12. 4 | 21. 18 |
| 4. 6 | 13. 3 | 22. 5 |
| 5. 6 | 14. 8 | 23. 8 |
| 6. 8 | 15. 4 | 24. 3 |
| 7. 7 | 16. 2 | 25. 11 |
| 8. 2 | 17. 3 | 26. 7 |
| 9. 7 | 18. 5 | |

Skill drill 2 (Medium basics)

- | | | |
|-------|--------|-----------------------------------|
| 1. 36 | 9. 33 | 17. 17 |
| 2. 32 | 10. 34 | 18. 14 and 15 |
| 3. 31 | 11. 24 | 19. 15 |
| 4. 25 | 12. 33 | 20. 12 |
| 5. 29 | 13. 37 | 21. 27 |
| 6. 26 | 14. 30 | 22. 7 |
| 7. 31 | 15. 22 | 23. 27 |
| 8. 13 | 16. 29 | 24. most values are from 12 to 18 |

Extension

- 18.8%
- still 1

Exit ticket

- Level 1: a graph with one dot for each value on a number line
- Level 2: 1
- Level 3: 8

10.04 Stem-and-leaf plots (booklet p. 27)

WE DO examples (teacher)

- Ex 1a: 25
- Ex 1b: 71 kg
- Ex 2a: 80 kg
- Ex 2b: 106 kg
- Ex 3a: 7 | 1 4 5 6 8 9 9; 8 | 0 0 0 2 2 3 4 5 8; 9 | 0 0 2 2 4 8; 10 | 4 5 6
- Ex 3b: to see the shape and find the median and mode easily

Easy · Set A

- 25
- 71 kg
- 80 kg
- 106 kg

Easy · Set A2 (quick drill)

- 27
- 50
- 19
- 83
- 104
- 66

Easy · Set B

- 15
- 45
- 75
- 66

Medium · Set C

- 2 | 1 3 7 8; 3 | 0 2 5 6 9; 4 | 1 4 5
- 6 | 4 7 8; 7 | 0 3 5 8 9; 8 | 2 5 8; 9 | 1

Challenging · Set D

- 7 | 1 4 5 6 8 9 9; 8 | 0 0 0 2 2 3 4 5 8; 9 | 0 0 2 2 4 8; 10 | 4 5 6
- 9 players, 9/25

Skill drill 1 (Easy basics)

- | | | |
|-------|--------|--------|
| 1. 47 | 8. 79 | 15. 56 |
| 2. 13 | 9. 66 | 16. 94 |
| 3. 11 | 10. 41 | 17. 92 |
| 4. 39 | 11. 87 | 18. 15 |
| 5. 12 | 12. 89 | 19. 32 |
| 6. 65 | 13. 82 | 20. 81 |
| 7. 27 | 14. 29 | 21. 37 |

22. 59	25. 16	28. 3	19. 9	25. 10	31. 1
23. 26	26. 21	29. 9	20. 9	26. 12	32. 3
24. 52	27. 65	30. 44	21. 12	27. 12	33. 10
			22. 11	28. 4	34. 9
			23. 11	29. 12	35. 1
			24. 4	30. 11	36. 8

Skill drill 2 (Medium basics)

1. 12, 15, 15, 18	10. 80, 82, 88, 89	16. 1 2 7 7 8; 2 ; 3
2. 31, 31, 33, 38	11. 60, 62, 64, 65	9; 4 5 6
3. 74, 74, 78, 79	12. 90, 94, 95, 99	17. 1 2; 2 5 6; 3 3
4. 21, 24, 26, 29	13. 1 7; 2 1 3 4; 3	5 9 9
5. 50, 51, 52, 56	; 4 3 8 9	18. 1 3 7; 2 2; 3 3
6. 15, 15, 18, 19	14. 1 6 9; 2 6; 3 3	5 6; 4 2
7. 30, 31, 34, 36	4; 4 7 9	19. 1 6 6; 2 2; 3 5
8. 11, 12, 14, 14	15. 1 6 7 8 8; 2 ; 3	6 8 9
9. 55, 55, 58, 59	4; 4 4 6	20. 1 4; 2 0 1; 3 5
		9; 4 2 7

Extension

1. 10 | 5; 11 | 2 8; 12 | 1 5; 13 | 0
2. 25 hundredths (0.25)

Exit ticket

1. Level 1: 6
2. Level 2: 40, 42, 47
3. Level 3: 1 | 2 4 7; 2 | 1 5 9

10.05 Mean and mode (booklet p. 35)

WE DO examples (teacher)

1. Ex 1a: 4
2. Ex 1b: 4
3. Ex 2a: 8.78
4. Ex 2b: 6
5. Ex 3a: 3.5; 1
6. Ex 3b: 6

Easy · Set A (quick drill)

1. 4	5. 12	9. 5
2. 6	6. red	10. 20
3. 1	7. 4	11. 5
4. 7	8. 3	12. 5

Easy · Set B

1. 5
2. 4
3. 30
4. 5
5. 3.5
6. 9

Medium · Set C

1. 8.78; 6
2. 3.5; 1
3. 14; 15
4. 4.07; 3.5
5. 49.86; 52
6. 1.88; 2

Challenging · Set D

1. 6
2. 84
3. e.g. 8, 8, 10, 11, 13
4. yes, e.g. 1, 1, 2, 2 (modes 1 and 2)

Skill drill 1 (Easy basics)

1. 3	7. 15	13. 12
2. 11	8. 9	14. 10
3. 6	9. 10	15. 7
4. 10	10. 16	16. 10
5. 10	11. 8	17. 10
6. 7	12. 13	18. 13

Skill drill 2 (Medium basics)

1. 31.4	9. 33.5	17. 20
2. 19.3	10. 29	18. 8
3. 30.7	11. 12.3	19. 8
4. 27.6	12. 32	20. 23
5. 27.3	13. 40	21. 28
6. 38.5	14. 53	22. 11
7. 28.1	15. 6	23. 10
8. 23	16. 24	24. 5

Extension

1. 10.4
2. it is a balance point, e.g. mean of 1 and 2 is 1.5

Exit ticket

1. Level 1: 5
2. Level 2: 8
3. Level 3: 6.5

10.06 Median and range (booklet p. 43)

WE DO examples (teacher)

1. Ex 1a: 4
2. Ex 1b: 4
3. Ex 2a: 10
4. Ex 2b: 6
5. Ex 3a: 3.5; 7
6. Ex 3b: the mean

Easy · Set A (quick drill)

1. 4	5. 13	9. 7
2. 6	6. 8	10. 3
3. 7	7. 3	11. 30
4. 19	8. 5	12. 6.5

Easy · Set B

1. 8
2. 4
3. 5
4. 7
5. 6
6. 22.5

Medium · Set C

1. 10; 6
2. 3.5; 7
3. 14.5; 6
4. 3.85; 1.6
5. 52; 23
6. 4; 98

Challenging · Set D

1. mean 7, median 3.5, mode 3, range 23
2. the median: the outlier 25 pulls the mean up
3. e.g. 1, 4, 6, 8, 11
4. 57

Skill drill 1 (Easy basics)

1. 10	5. 11	9. 13
2. 12	6. 22	10. 21
3. 15	7. 11	11. 8
4. 12	8. 20	12. 19

13. 19	21. 19	29. 35
14. 17	22. 46	30. 24
15. 26	23. 47	31. 32
16. 12	24. 27	32. 44
17. 13	25. 45	33. 35
18. 12	26. 29	34. 41
19. 4	27. 41	35. 37
20. 19	28. 29	36. 46

Skill drill 2 (Medium basics)

1. 23.5	14. 20.5	21. median 47, range 50
2. 23	15. 6	22. median 45, range 30
3. 17.5	16. median 22, range 43	23. median 35, range 40
4. 23	17. median 23, range 38	24. median 33, range 19
5. 15	18. median 29.5, range 40	25. median 47.5, range 31
6. 21	19. median 37, range 30	26. median 41, range 25
7. 18	20. median 43, range 32	27. median 36.5, range 46
8. 14.5		
9. 19		
10. 13		
11. 28.5		
12. 15		
13. 26.5		

Extension

- 3, 5, 9, 9
- a few very high salaries make the mean misleading

Exit ticket

- Level 1: 6
- Level 2: 25
- Level 3: 9

10.07 Analysing dot plots and stem-and-leaf plots (booklet p. 51)

WE DO examples (teacher)

- Ex 1a: 7
- Ex 1b: 1.5
- Ex 2a: 66
- Ex 2b: 64
- Ex 3a: 61.2
- Ex 3b: mostly in the 60s

Easy · Set A

- 7
- 1.5
- 1
- 1.88

Easy · Set A2 (quick drill)

- 6
- 16
- 0
- 20
- 8
- 18

Easy · Set B

- 30
- 64
- 66
- 61.2

Medium · Set C

- 13
- 23
- 23

- 23.2
- 23.48
- the range

Challenging · Set D

- 35
- 83
- 80
- 85.88

Skill drill 1 (Easy basics)

1. 40	7. 40	13. 11
2. 27	8. 41	14. no mode
3. 25	9. 51	15. 55
4. 35	10. 22	16. 40
5. 43	11. 44	17. 51.5
6. 47	12. 35	18. 5

Skill drill 2 (Medium basics)

1. 14	7. 5	12. 23
2. 9	8. no (it becomes 7)	13. 26
3. 6.5		14. 31
4. 8	9. 25	15. 46
5. 6.3	10. 33	16. 31
6. 1	11. 36	17. 42

Extension

- the median: it is not affected by extreme values
- 1.53

Exit ticket

- Level 1: 3
- Level 2: 53.5
- Level 3: 18

10.08 Comparing data (booklet p. 59)

WE DO examples (teacher)

- Ex 1a: the marks of the girls (left) and boys (right)
- Ex 1b: 94, a girl
- Ex 2a: 54
- Ex 2b: 83
- Ex 3a: girls: mean 55, median 54; boys: mean 55.8, median 60
- Ex 3b: girls: range 76 vs boys 83

Easy · Set A

- 94, girl
- 10, boy
- 8
- 13

Easy · Set A2 (quick drill)

- A
- B
- A
- B

Easy · Set B

- 55
- 54
- 76
- 55.8
- 60
- 83

Medium · Set C

- Class A: 5 | 8 2, 4 | 5 ...; Class B on the right
- Class A: higher median, smaller range

Challenging · Set D

1. boys have a slightly higher median (60 vs 54) but a larger range
2. girls: smaller range (76 vs 83)
3. median, as there are some very low marks
4. own questions

Skill drill 1 (Easy basics)

- | | | |
|-------|----------------------|---------------------|
| 1. A | 13. A | 22. A (15.25 and |
| 2. B | 14. A | 5.75) |
| 3. A | 15. B | 23. A (13.25 and |
| 4. B | 16. A | 9.5) |
| 5. B | 17. B | 24. A (14 and |
| 6. B | 18. A | 10.75) |
| 7. A | 19. A (13 and 10.75) | 25. B (9.25 and |
| 8. B | 20. A (8.75 and | 11.25) |
| 9. B | 7.25) | 26. B (14 and 15.5) |
| 10. B | 21. B (6.75 and | 27. A (7.75 and |
| 11. B | 10.25) | 5.25) |
| 12. B | | 28. A (11.75 and |
| | | 11.25) |

Skill drill 2 (Medium basics)

- | | | |
|--|--|---|
| 1. A mean 11,
range 4; B
mean 11.2,
range 9 | 4. A mean 13.4,
range 13; B
mean 13.6,
range 11 | 7. A 49, B 36: A
8. A 24, B 21: A
9. A 27, B 28: B
10. A 28, B 34: B
11. A 51, B 36: A
12. A 46, B 43: A |
| 2. A mean 11.8,
range 13; B
mean 11.6,
range 6 | 5. A mean 9.2,
range 8; B
mean 11.4,
range 11 | |
| 3. A mean 13.6,
range 13; B
mean 11.4,
range 13 | 6. A mean 13,
range 8; B
mean 10.8,
range 8 | |

Extension

1. Ana: she scores 12 or more every game
2. e.g. 5, 5, 5 and 0, 5, 10

Exit ticket

1. Level 1: comparing two groups
2. Level 2: A
3. Level 3: A